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Gujarat Section

DisasterShield: AI-Powered Multi-Hazard Early Warning System with Edge Intelligence

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GitHub: github.com/HarshParmar029/DisasterShield-SCAI2026



ABSTRACT

Climate change is significantly increasing the frequency and severity of natural disasters such as floods, wildfires and droughts across India, resulting in heavy loss of life, crops and property. Existing early-warning systems are largely centralized, expensive and too slow to provide the localized real-time alerts that vulnerable rural communities and farmers urgently need.

DisasterShield is a low-cost, edge-AI based multi-hazard early warning system that intelligently fuses real-time computer vision with multi-sensor data. A MobileNetV2 convolutional neural network trained using transfer learning achieves 97.99% validation accuracy for fire and smoke detection from a live camera feed. The vision confidence score is combined with five simulated environmental sensors (temperature, humidity, smoke level, rainfall and soil moisture) through a rule-based fusion engine that generates a single actionable risk level — Low, Medium or High.

The complete system is displayed on a live Flask web dashboard. The trained model has been successfully exported to TensorFlow Lite (9.2 MB), making it ready for future deployment on edge devices such as Raspberry Pi. The system is fully functional, offline-capable and completely open-source.

KEY CONTRIBUTIONS

- Real-time AI vision for fire & smoke detection
- Multi-sensor data fusion for multi-hazard risk
- Edge-ready using TensorFlow Lite (9.2 MB)
- Live Flask dashboard for instant alerts
- Fully functional, offline-capable & open-source

1. INTRODUCTION

Natural disasters such as floods, wildfires and droughts continue to cause significant loss of human life and agricultural damage across India. Traditional early-warning systems depend on centralized infrastructure and often fail to deliver fast, localized alerts to the rural communities and farmers who need them most.

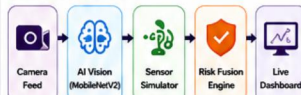
DisasterShield addresses this critical gap by combining four powerful components:

- Real-time AI Vision**
Accurate fire & smoke detection using MobileNetV2
- Multi-sensor Data Fusion**
Comprehensive multi-hazard risk assessment (Fire, Flood, Drought)
- Edge-ready Design**
TensorFlow Lite for offline and low-power operation
- Live Interactive Dashboard**
Real-time visualization of risk level and sensor values

This project is fully aligned with the SCAI 2026 theme "Connected Intelligence" and aims to deliver an affordable, practical and scalable early-warning solution that can genuinely help vulnerable communities.

2. METHODS

SYSTEM PIPELINE

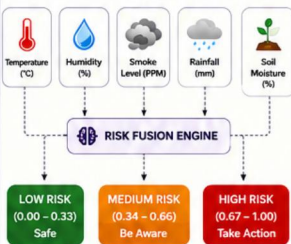


AI MODEL DETAILS

- Dataset:** Kaggle Fire Dataset containing 755 fire images and 244 non-fire images
- Architecture:** MobileNetV2 pretrained on ImageNet with a custom classification head
- Training Strategy:** 10 epochs of transfer learning with data augmentation (rotation, zoom and horizontal flip)
- Model Export:** Successfully converted to TensorFlow Lite format (9.2 MB) for lightweight edge inference

RISK FUSION LOGIC

The system combines the AI vision confidence score with five simulated sensor readings — Temperature, Humidity, Smoke Level, Rainfall and Soil Moisture. A carefully designed rule-based engine calculates a single risk score and maps it to three clear levels: **LOW**, **MEDIUM** or **HIGH**.



DASHBOARD IMPLEMENTATION

A Flask-based web application continuously streams the annotated live camera feed along with the current risk level and sensor values. The dashboard can be accessed from any browser on the local network.

TECHNOLOGY STACK



EDGE DEPLOYMENT READY



SCAN FOR GITHUB REPOSITORY



- Source Code
- Trained Model
- Detection Info
- Installation Guide
- Live Demo

3. RESULTS

97.99%

VALIDATION ACCURACY

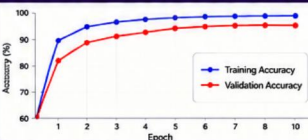
METRIC	VALUE
Training Accuracy	~99%
Validation Accuracy	96.98% – 97.99%
Model Size (TFLite)	9.2 MB
Detection Latency	< 1 second / frame

- Real-time fire and smoke detection is fully working on live camera feed
- Multi-hazard risk assessment successfully covers Fire, Flood and Drought signals
- Complete open-source system including training notebook and trained model
- Model is edge-ready and prepared for deployment on Raspberry Pi

LIVE DASHBOARD (SCREENSHOT)



TRAINING vs VALIDATION ACCURACY (10 EPOCHS)



4. CONCLUSIONS

DisasterShield successfully demonstrates that a practical, low-cost and accurate multi-hazard early warning system can be built using modern edge AI and intelligent sensor fusion. The MobileNetV2 model achieves strong validation accuracy of up to 97.99% and has been exported to a compact TensorFlow Lite format, making real-world edge deployment feasible.

FUTURE WORK

- Integration of real hardware sensors using Raspberry Pi with DHT22 and MQ-135
- Addition of LoRaWAN for long-range and low-power communication
- Incorporation of remote sensing and NDVI data for wider geographical coverage

The system has strong potential for real societal impact by enabling faster and more localized disaster alerts for vulnerable rural communities and farmers across India.

REFERENCES

- Kaggle Fire Dataset — [phyllake1337/fire-dataset](https://www.kaggle.com/datasets/phyllake1337/fire-dataset)
- Sandler et al., "MobileNetV2," CVPR 2018
- TensorFlow Lite Documentation, [tensorflow.org/lite](https://www.tensorflow.org/lite)
- SCAI 2026 — IEEE Student Branch, SVNIT Surat

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